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Lab 8
CS 455 Computer Networks
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I. Your Own Subnet In Binary

Interface Data:

Address: 10.128.25.1
Mask: 255.255.192.0 (0xffffc000)

Binary Alignment:

```
Address:    00001010.10000000.00|011001.00000001
Mask:       11111111.11111111.11|000000.00000000
            ———————————————-— + ———————————————
Network:    00001010.10000000.00|000000.00000000
Broadcast:  00001010.10000000.00|111111.11111111
```

Derived Values:

Prefix Length: /18
Network Address: 10.128.0.0
First Usable Host: 10.128.0.1
Last Usable Host: 10.128.63.254
Broadcast Address: 10.128.63.255

II. The Scripts Verdict on it

The lab8-subnet-tour.sh script agreed completely with these calculations.

III. The Five Cases

#	Prefix Length	Network Address	First Usable	Last Usable	Broadcast Address	Usable Host Count	Power of 2
1	/26	192.168.20.128	192.168.20.129	192.168.20.190	192.168.20.191	62	2^6. -2
2	/25	198.51.100.0	198.51.100.1	198.51.100.126	198.51.100.127	126	2^7. -2
3	/21	172.19.0.0	172.19.0.1	172.19.7.254	172.19.7.255	2046	2^11 -2

#	Prefix Length	Network Address	First Usable	Last Usable	Broadcast Address	Usable Host Count	Power of 2
4	/29	203.0.113.200	203.0.113.201	203.0.113.206	203.0.113.207	6	2^3 -2
5	/10	10.64.0.0	10.64.0.1	10.127.255.254	10.127.255.255	4,194,302	2^22 - 2

IV. My Error Report On The Five

All five cases were correct on the first try.

V. The Address That Is Its Own First Usable Host

Case 4 203.0.113.201/29 is the case where it is the first usable host in its block. In a /29 subnet, the last 3 bits are host bits. The octet value 201 in binary is 11001001, meaning the host portion is 001 (decimal 1). Since the host portion equals 1, it is easily identifiable that it is the first usable host.

VI. Five Same Or Different Predictions

1) 192.168.20.130 and 192.168.20.190 on subnet /26
   - My Guess: Same
   - Script Response: Same
   - Reduced Network Address: 192.168.20.128
2) 192.168.20.130 and 192.168.20.194 on subnet /26
   - My Guess: Different
   - Script Response: Different
   - Reduced Network Address: 192.168.20.128 (Host A) / 192.168.20.192 (Host B)
3) 172.19.5.77 and 172.19.2.9 on subnet /21
   - My Guess: Same
   - Script Response: Same
   - Reduced Network Address: 172.19.0.0
4) 10.62.0.19 and 10.99.4.1 on subnet /10
   - My Guess: Same
   - Script Response: Same
   - Reduced Network Address: 10.64.0.0

VII. My Kernels Two Answers

```
% route -n get 8.8.8.8
   route to: 8.8.8.8
destination: default
       mask: default
    gateway: 10.128.1.1
  interface: en0
```

```
      flags: <UP,GATEWAY,DONE,STATIC,PRCLONING,GLOBAL>
 recvpipe  sendpipe  ssthresh  rtt,msec    rttvar  hopcount      mtu
expire
       0         0         0         0         0         0      1500
0



% route -n get 10.128.25.50
   route to: 10.128.25.50
destination: 10.128.0.0
       mask: 255.255.192.0
  interface: en0
      flags: <UP,DONE,CLONING,STATIC>
 recvpipe  sendpipe  ssthresh  rtt,msec    rttvar  hopcount      mtu
expire
       0         0         0         0         0         0      1500
-1110
patrickfoulis@Patricks-MacBook-Air-4 Lab_8 %
```

When the kernel applies my local subnet mask to the local destination address, the resulting network address is 10.128.0.0, which matches my interface's assigned subnet. When the kernel applies the same mask to the remote destination, the bitwise and calculation gives 8.0.0.0, which does not match my local network address.

The destination 10.128.25.50 resides on the same Layer 2 broadcast domain as my machine, allowing direct frame delivery via ethernet without routing. The destination 8.8.8.8 resides outside of my local subnet, so traffic cannot be delivered directly. The kernel must forward packets to a default gateway (10.128.1.1) to carry traffic off the local network.

VIII. The Two Mask Drawing

Two-Mask Drawing (Question 1):

Mask 1 (/16): 11111111.11111111 | 00000000.00000000
Mask 2 (/20): 11111111.11111111 | 11110000.00000000

```
                  ------------------+-----+------------
                Handed Network | Sub | Host Bits
                       (16 bits) | net  | (12 bits)
                                 | (n=4)
```

Subnet bits borrowed (n): 4 bits (20 - 16 = 4)
Host bits remaining (h): 12 bits (32 - 20 = 12)

IX. The Five Counting Questions

1.Number of subnets and usable hosts per subnet:

Subnets: 2^4 = 16 subnets

Usable hosts: 2^12 - 2 = 4,094 usable hosts per subnet

2. Subnets from a /12 parent block:

Subnets: Borrowed bits n = 20 - 12 = 8 bits. 2^8 = 256 subnets.

Host count explanation: The usable host count did not change because host capacity depends on the remaining host bits to the right of the /20 boundary (32 - 20 = 12 bits), which stays 12 bits regardless of where the parent prefix starts.

3. Prefix for at least 300 usable addresses:

Prefix needed: /23

Arithmetic: Needs >= 300 hosts. 2^8 - 2 = 254 (too small); 2^9 - 2 = 510 usable hosts. Host bits needed = 9. Prefix = 32 - 9 = /23.

Wasted addresses: 512 total addresses - (300 needed + 2 reserved) = 210 addresses wasted.

4. Prefix for exactly 6 usable addresses:
Prefix needed: /29

Arithmetic: Needs 6 usable hosts + 2 reserved = 8 total addresses. 2^3 = 8. Host bits needed = 3. Prefix = 32 - 3 = /29.

Wasted addresses: 0 addresses wasted (8 total - 6 usable - 2 reserved = 0).

5. Assumption for "sixteen subnets from a /28":
It assumes the /28 block was created by subdividing a /24 parent block (2^4 = 16 subnets).
The starting parent block size, relative to which of the subnets were being calculated was not clarified.

X. The Address Plan

Segment Name	Hosts Needed	Prefix Chosen	Network Address	First Usable Host	Last Usable Host	Broadcast Address	Addresses Wasted
Students	500	/23	172.16.8.0	172.16.8.1	172.16.9.254	172.16.9.255	10
Faculty	200	/24	172.16.10.0	172.16.10.1	172.16.10.254	172.16.10.255	54
Staff	100	/25	172.16.11.0	172.16.11.1	172.16.11.126	172.16.11.127	26
Teaching lab	50	/26	172.16.11.128	172.16.11.129	172.16.11.189	172.16.11.191	12
Voice handsets	25	/27	172.16.11.192	172.16.11.193	172.16.11.222	172.16.11.223	5
Router link 1	2	/30	172.16.11.224	172.16.11.225	172.16.11.226	172.16.11.227	0
Router link 2	2	/30	172.16.11.228	172.16.11.229	172.16.11.230	172.16.11.231	0
Router link 3	2	/30	172.16.11.232	172.16.11.233	172.16.11.234	172.16.11.235	0

Unassigned Space Details: First Address Unassigned: 172.16.11.236
Addresses Remaining in /22 Block: 20 addresses (172.16.11.236 through 172.16.11.255)

XI. The Plan Done Backwards

Question 1:
Allocation Steps in Smallest-First Order:
* Router link 1 (/30): 172.16.8.0 - 172.16.8.3
* * Router link 2 (/30): 172.16.8.4 - 172.16.8.7
* * Router link 3 (/30): 172.16.8.8 - 172.16.8.11
* * Voice handsets (/27): 172.16.8.32 - 172.16.8.63
* * Teaching lab (/26): 172.16.8.64 - 172.16.8.127
* * Staff (/25): 172.16.8.128 - 172.16.8.255
* * Faculty (/24): 172.16.9.0 - 172.16.9.255
* * Students (/23): FAILS HERE.

Where and why it broke: Allocating from smallest to largest consumes the beginning of the space with small blocks, pushing the next available address for Students to 172.16.10.0. A /23 subnet requires 512 continuous addresses aligned on a 512-address boundary. Although 172.16.10.0 is

properly aligned, allocating a /23 spanning 172.16.10.0 through 172.16.11.255 leaves fragmented address gaps. Because of the addresses that were stranded beforehand.

XII. The One Size Comparison

Question 2:
Sizing for largest segment requires a /23. The total addresses needed for 8 segments using uniform sizing is 8 segments * 512 addresses = 4,096 total addresses (/20 block required).
A /22 block provides exactly 1,024 total addresses. Using uniform prefix sizing would require 4,096 addresses, which is 4 times the total capacity of the /22 block that was given.

Question 3:
Router link 1 immediately follows the Voice handsets segment. To double the Voice segment to a /26, Router link 1 and all subsequent router link blocks would have to be moved out of the way. Thus we can conclude that segments that are expected to grow should be placed either at the very end of the address allocations or adjacent to unassigned buffer space, rather than sandwiched between other fixed allocations.

XIII. The Shortcut

Case 2: 198.51.100.14 with 255.255.255.128
The mask boundary falls in the 4th octet (mask value m = 128).
Block size: 256 - 128 = 128. - Subnet multiples in 4th octet: 0, 128.
Destination octet value is 14, which falls in the block starting at 0.
Network Address: 198.51.100.0 (Matches Section 2).

Case 3: 172.19.5.77 with 255.255.248.0
The mask boundary falls in the 3rd octet (mask value m = 248).
Block size: 256 - 248 = 8.
Subnet multiples in 3rd octet: 0, 8, 16, 24, ...
Destination octet value is 5, which falls in the block starting at 0.
Network Address: 172.19.0.0 (Matches Section 2).

Case 4: 203.0.113.201 with 255.255.255.248
The mask boundary falls in the 4th octet (mask value m = 248).
Block size: 256 - 248 = 8.
Subnet multiples in 4th octet: 0, 8, 16, ..., 192, 200, 208.
Destination octet value is 201, which falls in the block starting at 200.
Network Address: 203.0.113.200 (Matches Section 2).

XIV. Legal and Illegal Masks

1. 255.255.244.0

Illegal, the third octet was the offending one: 11110100. Bit 6 contains zeros before the final ones, violating the continuous run of 1s required for a valid subnet mask.

2. 255.255.252.0

Legal

3. 255.254.0.0

Legal

4. 255.0.255.0

Illegal, the second octet was the offending one: 00000000. The entire 2nd octet consists of 8 zeros followed by ones in the 3rd octet (11111111), placing 0s before subsequent 1s.

5. 255.255.255.240

Legal

XV. The Broadcast Domain

Question 1:

There were 12 distinct source addresses. One out of the 12 of these addresses are on my subnet. Source Address Evaluated: 10.128.52.18, Local Subnet Range: 10.0.0.0 – 10.0.3.255. This address is outside of the range thus it is outside of the subnet.

Question 2:

The target IP address was 10.128.52.183, the requesting IP was 10.128.1.1, the sender's MAC address was Fortinet_09:00:20, and the destination MAC address was ce:90:bd:b6:82:1a. The fact that this frame arrived at my machine indicates that my host sits on the same Layer 2 broadcast domain / physical Ethernet switch segment as the Fortinet gateway. Because ARP requests are sent as Layer 2 broadcast frames, every device attached to that shared Layer 2 segment receives the frame, even if my machine was on a different IP subnet or prefix than 10.128.0.0/16.

Question 3:

When I attempted to ping I got a timeout request as the lab said would be expected out of some OS and after further research this is normal behavior for macOS.

XVI. Two Subnets One Switch

No the two machines cannot reach each other directly at layer 3 because their IP addresses are on different subnets. This forces the machines to route to a default gateway rather than resolving the destination locally via the ARP. However, if one machine broadcasts, the frame will arrive at the other machine because an unmanaged switch operates purely at Layer 2 and does not know about IP addresses or subnets. The switch simply floods any MAC destination out of all of the active physical ports. To enforce isolation you would need to add a VLAN

XVII. Specification Questions

Part A — The Original Procedure (RFC 950)

1. "The motivation for sub-netting is to provide a multi-level hierarchical routing structure, and thereby alleviate routing table growth."Yes, this reason still applies. Aggregating all 8 internal segments under the single /22 block allows external routers to store only one routing entry for the entire organization instead of individual routes for every subnet.

2. "It is not required that the 1-bits in a subnet mask be contiguous.".When given 255.255.244.0, the script rejected it with an error stating that it is not a legal mask because it contains a 0 before its last 1. Current practice moved to strictly contiguous masks because non-contiguous masks break standard prefix length notation and significantly increase processing complexity in modern router forwarding lookup hardware.

3. ICMP Address Mask Request (Type 17) and ICMP Address Mask Reply (Type 18).RFC 6918 formally retired these ICMP messages, marking them "Historic". Modern hosts obtain their subnet mask automatically via DHCP (Dynamic Host Configuration Protocol) or IPv6 Neighbor Discovery / Router Advertisements.

4. “In a point-to-point link, there are no other hosts on the link. Therefore, packets directed to the broadcast address... do not need to be broadcast to other hosts."A /31 network provides only 2 total IP addresses. An Ethernet segment with 3 hosts requires at least 3 distinct host IP addresses plus additional addresses for broadcast operations. A /31 cannot provide enough addresses for 3 devices and lacks the required broadcast capability.

5. "In a network with 500 point-to-point links, for example, this practice would amount to a savings of 1000 addresses (the equivalent of four class C address spaces)."
   - Savings for Section 5 Plan:
     - The Section 5 address plan contains 3 point-to-point router links:
     - Standard /30 allocation: 3 links x 4 addresses = 12 addresses used.
     - RFC 3021 /31 allocation: 3 links x 2 addresses = 6 addresses used.
     - Total Addresses Saved: 6 addresses returned to the available block.

XVIII. Reflection

Moving the subnet boundary one line to the left gives up half of the available addresses per subnets. Which prevents expansion of the IP addresses if it is later needed by local end users. However it gains the amount of available subnets, which gives network administrators larger flexibility.